CivicPulse

FYP Code: 23F-90

Software Requirement Specification

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NOVEMBER 2026

Document Revision History

Version	Date	Prepared by	Description
1.0	November 2026	Ayoush Kumar, Sarichand, Rainuka Bai	Initial version for supervisor review
1.1	November 2026	Ayoush Kumar, Sarichand, Rainuka Bai	Revised after supervisor review: account and session features, upload limit, testable performance targets, traceability to test cases, references and figures

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1. Introduction

1.1. Purpose of Document

This Software Requirements Specification defines the functions, interfaces and quality constraints of CivicPulse. It converts the approved proposal for project 23F-90 [12] into numbered requirements and use cases, against which the system can be built and assessed. The design that satisfies these requirements is described in the companion Software Design Specification.

1.2. Intended Audience

Table 1: Intended audience

Reader	Use of this document
Supervisor and co-supervisor	Confirm that the requirements agree with the approved proposal and resolve open decisions.
FYP evaluation committee	Assess the scope and compare the delivered system with the stated requirements.
Development team	Plan work from individual requirements and derive tests from them.
Future maintainers	Understand the required behaviour before extending the system.

1.3. Document Convention, abbreviations, and acronyms

Body text is set in Arial 11 point, headings in bold Arial and tables in Arial 9.5 to 10 point, and the running header in italic Times New Roman. Identifiers use fixed prefixes: FR for functional requirements, NFR for non-functional requirements, UC for use cases and O for project objectives. The word "shall" denotes a mandatory requirement, and "should" a target that is desirable but not mandatory. Bracketed numbers refer to the entries in Section 6.

Table 2: Terms, abbreviations and acronyms

Term	Definition
Administrator / Reviewer	The person operating the dashboard. A team member performs this role during the demonstration.
API	Application Programming Interface
EXIF	Exchangeable Image File Format, metadata within an image that can include a position
Gate	One of five sequential checks applied to a claimed repair
GiST	Generalized Search Tree, the index type used for spatial queries
JWT	JSON Web Token, a signed token identifying a logged-in user
mAP@0.5	Mean Average Precision at an overlap threshold of 0.5
ORB	Oriented FAST and Rotated BRIEF, a keypoint detection and matching method [8]
OTP	One-Time Password, here the 6-digit email code
Pin	A single physical hazard on the map, grouping one or more reports
RANSAC	Random Sample Consensus, used to reject inconsistent keypoint matches
Report	One citizen submission comprising a photograph, category, position and accuracy value
SMTP	Simple Mail Transfer Protocol
Supporter	A distinct account that has reported or confirmed a pin
YOLO	You Only Look Once, a family of object detection models [9], [14]

2. Overall System Description

2.1. Project Background

Road damage in Sindh worsens rapidly during the monsoon. Provincial figures for the 2022 floods record 8,463 km of damaged road, including 1,055 km in Sukkur Division [1]. Available funding covers only part of this network; a Rs 1.5 billion allocation for Karachi in 2024, for example, covered only a few selected roads [2]. Municipal engineers must therefore prioritise repairs, which requires reliable information about each defect.

Existing channels provide little of this information. The federal government reports that 98.7% of complaints on the Pakistan Citizen's Portal were resolved, yet only about 30% of complainants said they were satisfied [3]. Helplines create one ticket for each caller. Platforms such as FixMyStreet reduce duplicates mainly by publishing reports online for everyone to see [4]. Dashcam studies detect damage accurately, but require a survey vehicle and do not extend beyond detection [5]. An authority therefore cannot readily determine how many separate hazards exist, how severe they are or whether a repair has taken place.

To address these gaps, CivicPulse groups duplicate reports into a single pin, estimates severity from the photograph using a trained model, and accepts a repair only after photographic evidence has passed automatic checks and human review.

2.2. Project Scope

The project covers road hazards in Sukkur and delivers three connected products:

Android application: hazard map, photographic reporting with an offline queue, hazard confirmation, volunteer tasks, repair checks, points, leaderboard and coupons.

Server: report intake, secure image storage, merge engine, YOLO analysis in a background worker, five-gate repair pipeline, hazard ranking, weather warnings and reward rules.

Web dashboard: ranked queue, map, pin detail and split function, claim review, coupon management, audit log and summary statistics.

The team will also collect and label three datasets: local road photographs, a ground-truth set for merge evaluation and a set of deliberately faked repairs.

2.3. Not in Scope

Physical road repair, and any volunteer activity that would endanger a person, including pothole filling in live traffic.

Live municipal deployment. No agreement exists with Sukkur Municipal Corporation, so a team member will act as reviewer and results will be reported accordingly.

A contracted shop network. If no shop participates, coupons will be simulated and labelled as such.

Ranking weights learned from data, since no labelled ranking dataset is available.

Detection of one person operating several verified accounts. The per-pin cap and human gate limit the impact of this abuse but do not detect it.

Formal security auditing, navigation and prediction of future defects.

An iPhone build and cloud hosting, which remain optional after the defence.

2.4. Project Objectives

Table 3 lists the project objectives. Each has a defined measure, and all except the stretch objective O7 must be achieved for the defence.

Table 3: Project objectives and measures

ID	Objective	Measure
O1	Merge reports of one physical hazard into one pin using distance, category and time.	Precision and recall against a hand-labelled set, with a radius sweep from 5 m to 40 m using ST_ClusterDBSCAN [7], [13].
O2	Detect road-surface defects and estimate severity from a single photograph.	mAP@0.5 on unseen Sukkur photographs, for the public-data model and the fine-tuned model.
O3	Verify a claimed repair from a before-and-after pair of photographs.	False-accept and false-reject rates on faked and genuine repairs, with gates disabled and then enabled.
O4	Rank hazards using a transparent score.	Spearman's rho against independent rankings of about 50 hazards by three people.
O5	Warn users in an affected zone and raise the priority of weather-sensitive hazards.	Median delivery time and the proportion of notified users inside the zone.
O6	Award points, order the leaderboard and issue single-use coupons correctly.	An automated test suite, including simultaneous redemption of one coupon.
O7 (stretch)	Observe real use, subject to recruiting a user group.	Indicative figures from a two-week trial.

2.5. Stakeholders

Table 4: Stakeholders

Stakeholder	Role or interest
Citizens and volunteers	Report and confirm hazards, carry out safe tasks and earn rewards.
Municipal reviewer (performed by a team member)	Receives a ranked list with supporting evidence and approves repairs.
Sukkur Municipal Corporation (potential)	May obtain a measured record of hazards and repairs if the system is adopted.
Local shops (potential)	May provide coupons in return for visibility.
Dr. Saif Hassan and Dr. Ismail	Project supervision and approval of major decisions.

Team responsibilities. Ayoush Kumar: the detector, severity estimation, Gates 3 and 4, and the ground-truth and faked-repair datasets. Sarichand: accounts, reporting, image handling, Gates 1 and 2, ranking, the weather service and the radius sweep. Rainuka Bai: the map, dashboard, audit log, merge engine, points, leaderboard and coupons.

2.6. Operating Environment

Table 5: Operating environment

Component	Environment
Mobile application	Android phones with a camera and GPS, running a React Native build created with Expo. The application shall remain usable with weak or no coverage. The demonstration build is an APK made with EAS Build.
Dashboard	A current desktop web browser on a large screen.

Component	Environment
Server	A laptop or small server running Docker Compose, with Python 3.12, FastAPI, PostgreSQL 16 and PostGIS 3.4. YOLO runs on the CPU for the demonstration.
Model training	Free GPU sessions on Google Colab or Kaggle, separate from the operational system.
Network	Mobile data or Wi-Fi between clients and server. Internet access is required for forecasts, push messages and map tiles.

2.7. System Constraints

Software: only free and open tools may be used. The application is limited to the capabilities of Expo; remote push and the map library require a development build instead of Expo Go.

Hardware: no hardware will be purchased. Inference must run on an ordinary laptop, which favours a small YOLO variant.

Cultural: this version uses English, and public views show a chosen display name in place of a real name.

Legal: photographs and positions are personal data, so their exposure is minimised. Faces and number plates in collected datasets are blurred, and volunteer tasks must be lawful for private persons.

Environmental: use is outdoors, in sunlight and rain, near moving traffic. Screens therefore need large controls and few steps, and the application must not require operation while driving.

User: users have varied technical ability and may own low-end phones, so workflows must be short and error messages clear.

Off-the-shelf components: Open-Meteo terms apply to non-commercial use [10], public map tiles require attribution, and phone GPS is typically accurate to about 4.9 m in open sky and less accurate near buildings [6].

Academic: the work must fit the 26-week plan and be demonstrable at the defence.

2.8. Assumptions & Dependencies

Assumptions. The following working decisions apply until confirmed by the supervisor:

Verification uses an emailed code, as SMS incurs a cost. The proposal permits either a phone number or an email address.

Hazard categories are pothole, waterlogged road, blocked drain, fallen branch or debris, and other. The detector covers road-surface defects only; other categories use a default severity and administrator review.

Merge parameters, score weights, point values and weather thresholds in Appendix B are initial values, to be tuned using measured data.

A team member acts as municipal reviewer, and also as the shop when coupons are simulated.

In the demonstration the server runs on a team laptop on the phones' Wi-Fi network. Its address can change, so the user can set it in the application.

Dependencies. The system depends on Open-Meteo, the Expo push service, an SMTP provider and OpenStreetMap tiles. It also depends on the accuracy value and mock-location flag reported by the phone [11], the RDD2022 dataset [5] and timely collection of local data by the team.

3. External Interface Requirements

CivicPulse exchanges data with its users, the phone hardware and four external services, as shown in the context diagram below. Clients communicate only with the server API, except when downloading map tiles.

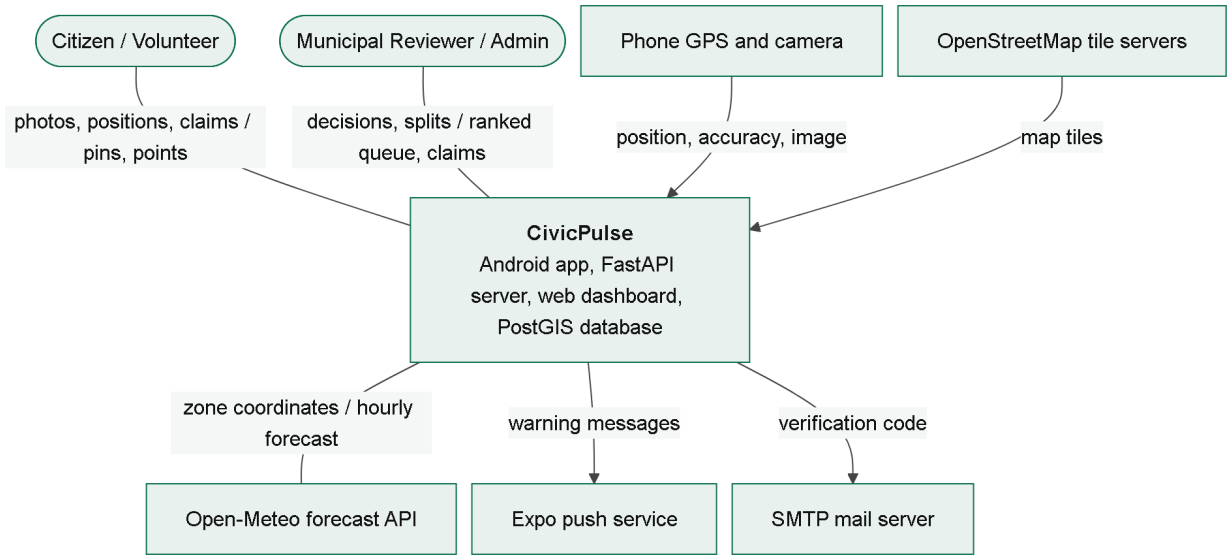


Figure 1: System context diagram

3.1. Hardware Interfaces

Table 6: Hardware interfaces

Device	Interaction
Phone camera	Captures photographs in the application through expo-camera. Gallery selection is not offered.
Phone GPS	Supplies latitude, longitude, accuracy in metres and a mock-location flag through expo-location, only while the application is open.
Phone storage	Holds queued reports and photographs until upload, and tokens in the secure keystore.
Server disk	Stores re-encoded photographs under random file names.
Desktop or laptop	Runs the dashboard in a web browser.

3.2. Software Interfaces

Table 7: Software interfaces

Software	Version	Purpose and data exchanged
PostgreSQL + PostGIS	16 / 3.4	Stores all records. Geography columns and spatial indexes answer distance queries.
FastAPI, SQLAlchemy, Alembic	0.162 / 2.1 / 1.20	Web API, database access and schema migrations.
React Native + Expo	SDK 58	Mobile runtime, secure storage, camera, location and push registration.
React + Vite	Current stable	Dashboard front end.
MapLibre + OpenStreetMap	Current stable	Map display on both clients, with tiles downloaded per viewport.

Software	Version	Purpose and data exchanged
Ultralytics YOLO [9]	Fixed in Phase 4	Receives an image and returns boxes, classes and confidence values.
OpenCV	Fixed in Phase 5	ORB features, matching and homography estimation for Gate 3.
Pillow	Current stable	Decodes and re-encodes uploads and removes metadata.
Open-Meteo API [10]	Public service	Receives zone coordinates and returns hourly rainfall, rain probability and wind gusts.
Expo push service	Public service	Receives device tokens and warning text and delivers the notification.

3.3. Communications Interfaces

Client to server: REST over HTTP. The demonstration build allows unencrypted HTTP so that it can reach the server laptop on a private network; any public build shall allow HTTPS only. Request bodies use JSON, except photograph uploads, which use multipart form data.

Authentication: a bearer JWT on each protected request, and an X-Device-Id header identifying the phone for rate limiting.

Status codes: 400 incorrect code, 401 not authenticated, 403 wrong role or unverified account, 409 conflict, 422 invalid input, 423 locked code, and 429 rate limited with a Retry-After header.

Email: SMTP, with STARTTLS where the provider supports it. Development email is captured by a local Mailpit inbox.

Synchronisation: the application retries uploads with increasing delays, and the server treats a repeated client identifier as the same report.

Media: photographs are served from an origin separate from the dashboard, with content sniffing disabled.

4. Functional Requirements

4.1. Functional Hierarchy

The system comprises nine functional areas, shown in the hierarchy below and specified in the requirements table that follows. Priority is Must for functions required at the defence, and Should for functions that may be reduced under the project's risk plan.

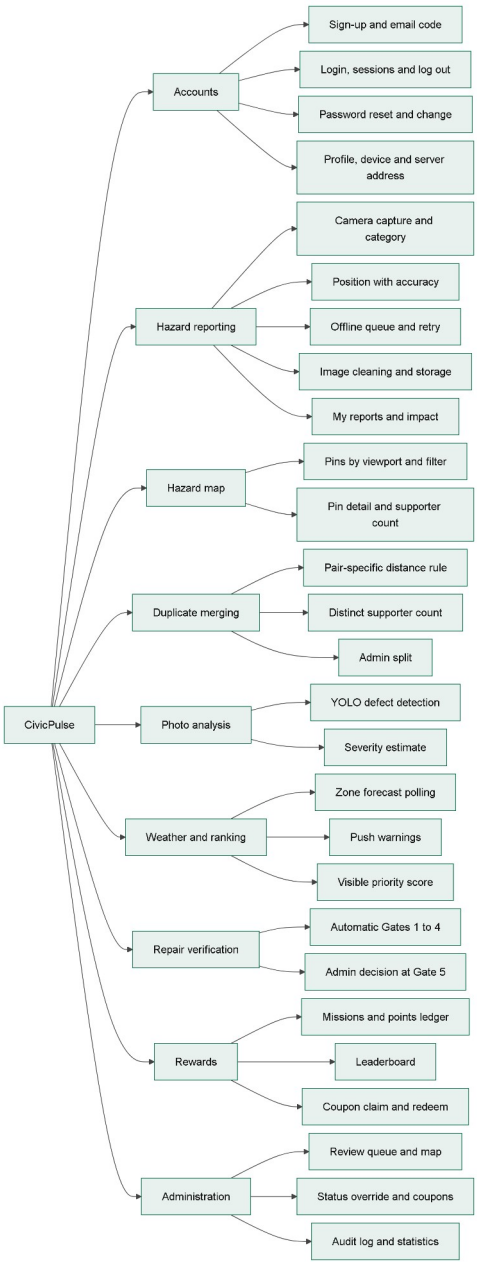


Figure 2: Functional hierarchy

Table 8: Functional requirements

ID	Requirement: the system shall	Priority
FR-01	Register a citizen with email, password and display name, and reject duplicate emails.	Must

ID	Requirement: the system shall	Priority
FR-02	Verify an email using a 6-digit code that expires in 10 minutes and locks after 5 incorrect attempts.	Must
FR-03	Issue a 15-minute access token and a 14-day refresh token at login, bound to a server-side session; rotate the refresh token on every use, accept the previous one for 60 seconds after a rotation, and revoke the session when an older token is reused.	Must
FR-04	Refuse protected actions from unverified or disabled accounts, and administrator actions from citizens.	Must
FR-05	Create administrator accounts only through a server-side script, never through the API.	Must
FR-06	Allow users to change their display name and register a device.	Must
FR-07	Capture a report containing an in-app photograph, category, optional note, position, accuracy, time and mock-location flag.	Must
FR-08	Queue reports on the phone when offline and upload each of them exactly once.	Must
FR-09	Verify uploads by content, limit their size, re-encode them, remove metadata, and store them under random names with a SHA-256 hash.	Must
FR-10	Display open pins for the visible map area with filters and supporter counts, and without reporter identity.	Must
FR-11	Attach a report to the nearest open pin of the same category inside the pair threshold and active window, and otherwise create a new pin.	Must
FR-12	Count supporters as distinct accounts and record each merge decision on its report.	Must
FR-13	Never add a report to a closed pin; instead, link a new pin to the closed one.	Must
FR-14	Allow an administrator to split a pin with no loss of reports.	Must
FR-15	Analyse each photograph with YOLO in the background, and store the boxes and model version.	Must
FR-16	Compute severity from 1 to 5, using a category default when nothing is detected.	Must
FR-17	Poll hourly forecasts for each zone, retain snapshots and continue operating after a failed call.	Must
FR-18	Send one push warning per event to devices in the affected zone, and record sent and received times.	Must
FR-19	Compute a priority score with a visible breakdown, and recompute it when its inputs change.	Must
FR-20	Store ranking weights as versioned settings and log each change.	Should
FR-21	Issue a one-time token when a repair mission is accepted, and refuse a checker who reported the pin.	Must
FR-22	Run Gates 1 to 4 in order, stop at the first failure and store all measurements.	Must
FR-23	Pay no reward before administrator approval at Gate 5, and allow approval of an automatically rejected claim.	Must
FR-24	Provide switches that disable Gates 2, 3 and 4 for the evaluation run.	Must
FR-25	Record points in an append-only ledger with a unique award key, and accept no point value from a client.	Must
FR-26	Hold report points until the pin has a second supporter or administrator approval, and enforce a per-pin cap.	Must
FR-27	Order the leaderboard by points using a stable tie rule.	Must
FR-28	Issue coupon codes against a balance, and redeem each code at most once, including under simultaneous requests.	Must
FR-29	Offer safe volunteer tasks, verified by before-and-after photographs and administrator approval.	Should
FR-30	Provide a ranked queue, map, pin detail, claim review, coupon management and statistics on the dashboard.	Must
FR-31	Write an audit record for each approval, rejection, override, merge correction and split, in the same transaction as the change.	Must

ID	Requirement: the system shall	Priority
FR-32	Refuse passwords shorter than 8 or longer than 128 characters, without an uppercase letter, a lowercase letter, a number and a symbol, found in a common-password list, or containing the email name, at sign-up and at every password change.	Must
FR-33	Reset a forgotten password with a 6-digit emailed code under the expiry and lock rules of FR-02, without revealing whether an email is registered, and end all of the account's sessions.	Must
FR-34	Let a logged-in user change the password after confirming the current one, keeping the current session and ending all others.	Must
FR-35	End the current session at logout, so that its refresh token is refused from then on.	Must
FR-36	Show each citizen their own reports, including those still queued on the phone, with the state of each report and the status of the hazard it joined.	Must
FR-37	Show each citizen server-computed totals of reports, hazards, confirmations by other people, hazards in repair and hazards fixed, with ranks and badges derived from those totals only.	Should
FR-38	Let the user set the server address in the application, check that a CivicPulse server answers there before saving it, and name the address in any could-not-reach message.	Should

4.2. Use Cases

The use case diagram below shows the primary and secondary actors and their use cases. A volunteer is a citizen who has accepted a mission, and therefore inherits all citizen use cases. External services participate as secondary actors.

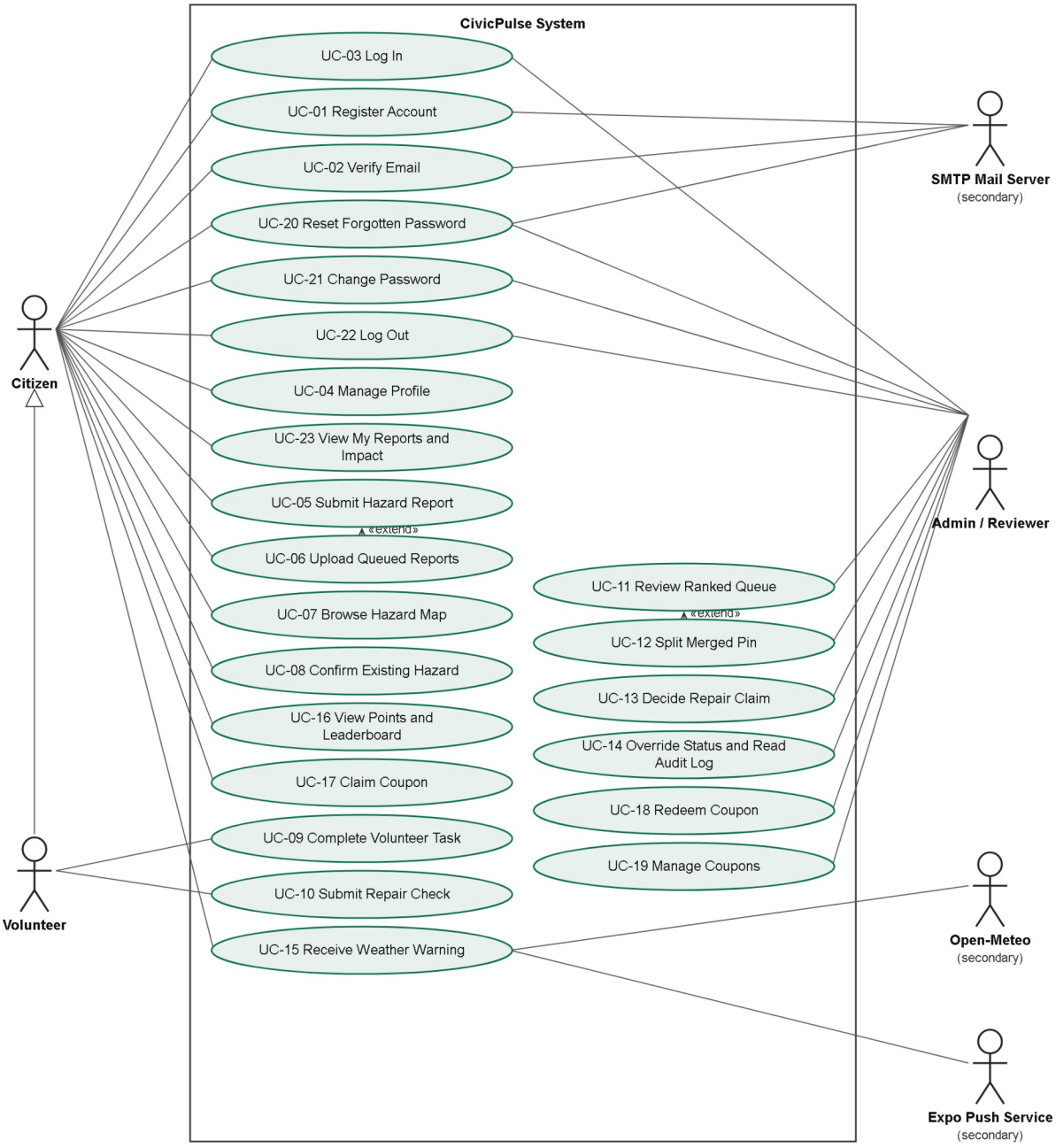


Figure 3: Use case diagram

4.2.1. UC-01: Register Account

A new user creates a citizen account, so that subsequent reports and rewards can be attributed to one person.

UC-01: Register Account		
Use case Id:	UC-01	
Actors: Citizen (initiator), SMTP mail server		
Feature: Accounts		
Pre-condition:	The app is installed and the phone can reach the server.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the sign-up screen and enters a display name, email and password.	Checks the format of each field on the phone.
2.	Taps Create account.	Counts the request against the device and network limits, hashes the password with Argon2 and saves the account as unverified.
3.	(System step)	Generates a 6-digit code, stores only its keyed hash, emails the code and opens the code entry screen.
Alternate Scenarios		
2a: The email already belongs to an account. The server answers 409 and the app suggests logging in. 2b: The password breaks the rule in FR-32, for example it has no symbol or contains the email name. The server answers 422 with the reason, and nothing is stored. 2c: Too many sign-ups came from this device. The server answers 429 with a retry time.		
Post Conditions		
Step#	**Description**	
1.	An unverified citizen account exists.	
2.	One active verification code exists for that account.	
Use Case Cross referenced	UC-02	

4.2.2. UC-02: Verify Email

The citizen confirms ownership of the email address by entering the code sent to it.

UC-02: Verify Email		
Use case Id:	UC-02	
Actors: Citizen		
Feature: Accounts		
Pre-condition:	An unverified account exists and a code has been sent.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Types the 6-digit code and taps Verify.	Locks the code record, then checks its expiry and attempt count.
2.	(System step)	Compares the code, marks it used and marks the account verified.
3.	(System step)	Returns the citizen to the login screen with a confirmation message.
Alternate Scenarios		
1a: The code is wrong. The attempt count rises and the citizen may try again. 1b: This is the fifth wrong try. The code locks and even the right value is refused (423). 1c: Ten minutes have passed. The server reports an expired code. 1d: The citizen taps Send a new code. The old code is cancelled and a new one is emailed, at most three times in ten minutes.		
Post Conditions		
Step#	**Description**	
1.	The account is verified and can log in.	
Use Case Cross referenced	UC-01, UC-03	

4.2.3. UC-03: Log In

A verified citizen or an administrator obtains access and refresh tokens. The phone retains the tokens across application restarts.

UC-03: Log In		
Use case Id:	UC-03	
Actors: Citizen or Admin		
Feature: Accounts		
Pre-condition:	The account exists, is verified and is not disabled.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Enters email and password and taps Log in.	Counts the attempt per account, per device and per network address.
2.	(System step)	Verifies the Argon2 hash and issues a 15-minute access token and a 14-day refresh token with the profile.
3.	(System step)	The app stores the tokens in SecureStore, registers the device and opens the main screen.
Alternate Scenarios		
1a: The credentials are wrong. A single generic message is shown, so nobody learns whether the email exists. 1b: The email is unverified. The app moves to the code entry screen. 1c: A sixth attempt within a minute receives 429. 3a: The access token expires later. The app silently requests a new one with the refresh token. If that is rejected, the citizen returns to login.		
Post Conditions		
Step#	**Description**	
1.	The client holds valid tokens.	
2.	The device record is current.	
Use Case Cross referenced	UC-02, UC-04	

4.2.4. UC-04: Manage Profile

The citizen views the account details and changes the name displayed on the leaderboard.

UC-04: Manage Profile		
Use case Id:	UC-04	
Actors: Citizen		
Feature: Accounts		
Pre-condition:	The citizen is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the profile screen.	Shows email, role and display name.
2.	Edits the display name and taps Save.	Validates the length, saves the new name and returns the updated profile.
Alternate Scenarios		
2a: The name is shorter than 2 or longer than 40 characters. The server answers 422. 2b: A request also tries to set a role. The field is ignored and the role does not change.		
Post Conditions		
Step#	**Description**	
1.	The new display name is used wherever the citizen is shown.	
Use Case Cross referenced	UC-03, UC-16	

4.2.5. UC-05: Submit Hazard Report

The citizen photographs a hazard on site and submits it with a category and position. This is the principal citizen function.

UC-05: Submit Hazard Report		
Use case Id:	UC-05	
Actors: Citizen		
Feature: Hazard reporting		
Pre-condition:	The citizen is logged in and has granted camera and foreground location permission.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the report screen and takes a photo with the in-app camera.	Records latitude, longitude, accuracy radius, capture time and the mock-location flag.
2.	Picks a category, optionally adds a note and taps Send.	Saves the report on the phone with a client-generated identifier and starts the upload.
3.	(System step)	The server cleans and stores the photo, then applies the merge rule.
4.	(System step)	The report is attached to a pin, a vision job is queued and the app shows the report status.
Alternate Scenarios		
1a: A permission was refused. The app explains why it is needed and how to allow it. 2a: There is no signal. The report stays queued and UC-06 takes over. 3a: The file is not a real image or exceeds 5 MB. The upload is rejected with a clear reason. 3b: The position is flagged as mocked. The report is kept but marked as untrusted.		
Post Conditions		
Step#	**Description**	
1.	The report and its image are stored permanently.	
2.	The report belongs to exactly one pin.	
3.	The pin's supporter count reflects distinct accounts.	
Use Case Cross referenced	UC-06, UC-07, UC-08	

4.2.6. UC-06: Upload Queued Reports

Reports created without network coverage are delivered automatically once a connection returns, and each is stored exactly once.

UC-06: Upload Queued Reports		
Use case Id:	UC-06	
Actors: Citizen (indirectly, by regaining coverage or reopening the app)		
Feature: Hazard reporting		
Pre-condition:	At least one report is waiting in the phone's local queue.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Regains a network connection or reopens the app.	Detects connectivity and sends the oldest queued report.
2.	(System step)	The server checks the client identifier and stores the report if it is new.
3.	(System step)	After the confirmation arrives, the app deletes the item from its queue and updates My Reports.
Alternate Scenarios		
1a: The upload fails. The app retries later with increasing delay and keeps the item. 2a: The server already holds this identifier, because an earlier confirmation was lost. It returns the existing report and creates no duplicate.		
Post Conditions		
Step#	**Description**	
1.	Each queued report exists once on the server.	
2.	The local queue no longer holds confirmed items.	
Use Case Cross referenced	UC-05	

4.2.7. UC-07: Browse Hazard Map

A logged-in user views the hazards in the visible map area. Reporter identities are not disclosed.

UC-07: Browse Hazard Map		
Use case Id:	UC-07	
Actors: Citizen		
Feature: Hazard map		
Pre-condition:	The citizen is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the map or moves it.	Requests pins inside the visible area and draws them with supporter badges.
2.	Applies a category or status filter.	Requests and draws the matching pins.
3.	Taps a pin.	Shows photo, category, severity, status and supporter count.
Alternate Scenarios		
1a: There is no connection. The app shows the last loaded pins with a notice. 1b: Closed pins are left out by default. A new pin at the same place links to the closed one it follows.		
Post Conditions		
Step#	**Description**	
1.	No data has changed.	
Use Case Cross referenced	UC-05, UC-08, UC-15	

4.2.8. UC-08: Confirm Existing Hazard

A second person adds supporting evidence to a hazard reported by someone else, which increases its corroboration count.

UC-08: Confirm Existing Hazard		
Use case Id:	UC-08	
Actors: Citizen		
Feature: Missions and rewards		
Pre-condition:	The citizen is logged in, stands near an open pin and did not report it.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Taps Confirm on the pin detail sheet.	Opens the camera for a fresh photo.
2.	Takes a photo and sends it.	Checks the position against the same pair-specific threshold used for merging.
3.	(System step)	Attaches the photo as a new report, recounts distinct supporters and awards the confirmation points once.
Alternate Scenarios		
2a: The citizen is too far away. The confirmation is refused and no points are awarded. 3a: The citizen reported or already confirmed this pin. The photo is stored as a report on the pin, but it earns no points and does not change the supporter count.		
Post Conditions		
Step#	**Description**	
1.	The supporter count is correct.	
2.	The original reporter's held points may now be released.	
Use Case Cross referenced	UC-05, UC-07, UC-16	

4.2.9. UC-09: Complete Volunteer Task

A volunteer carries out a lawful and safe task, such as clearing debris or unblocking a drain. Pothole filling is excluded.

UC-09: Complete Volunteer Task		
Use case Id:	UC-09	
Actors: Volunteer, Admin		
Feature: Missions and rewards		
Pre-condition:	The volunteer is logged in and a task is available on an open pin.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the mission list and accepts a task.	Records the mission and shows safety guidance.
2.	Takes before and after photos on site and submits them.	Applies the freshness and position checks.
3.	(System step)	Places the task in the admin's review list.
4.	(System step)	When the admin approves, awards the task points once.
Alternate Scenarios		
2a: A check fails. The volunteer sees the reason and no points are awarded. 4a: The admin rejects the task. The reason is logged and shown to the volunteer.		
Post Conditions		
Step#	**Description**	
1.	The mission has a final status.	
2.	Any award appears once in the ledger.	
Use Case Cross referenced	UC-13, UC-16	

4.2.10. UC-10: Submit Repair Check

A volunteer photographs a location that appears to have been repaired, and the server evaluates the photograph through the automatic gates in order of cost.

UC-10: Submit Repair Check		
Use case Id:	UC-10	
Actors: Volunteer		
Feature: Repair verification		
Pre-condition:	The pin is open or pending repair, and the volunteer is not its reporter.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Accepts the repair-check mission while online.	Issues a one-time token with an expiry and stores only its hash.
2.	Reads the framing tips and takes the after photo.	Queues the photo, so it can upload later if coverage is lost.
3.	(System step)	Gate 1: confirms the token and that the image hash is not already on this pin.
4.	(System step)	Gate 2: confirms the position lies inside the pin's threshold and notes any mock flag.
5.	(System step)	Gate 3: matches ORB keypoints with the before photo and requires enough RANSAC inliers.
6.	(System step)	Gate 4: maps the defect box into the after photo and runs YOLO on that patch only.
7.	(System step)	Saves all measurements and marks the claim as awaiting the admin.
Alternate Scenarios		
1a: The volunteer is the pin's reporter. The mission is refused. 3a to 6a: A gate fails. The run stops there, later gates are skipped, and the claim records the failure. The admin may still review it. 6b: The mapped box falls outside the frame or is too small. This counts as a Gate 4 failure.		
Post Conditions		
Step#	**Description**	
1.	A repair claim exists with every gate result.	
2.	No points have been paid.	
Use Case Cross referenced	UC-13, UC-16	

4.2.11. UC-11: Review Ranked Queue

The reviewer examines the ranked list of open hazards and the contribution of each factor to a hazard's position.

UC-11: Review Ranked Queue		
Use case Id:	UC-11	
Actors: Admin		
Feature: Administration		
Pre-condition:	The admin is logged in to the dashboard.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the queue.	Lists open pins by priority score, with ties broken in favour of the older pin.
2.	Selects a pin.	Shows severity, supporters, days open, weather risk and each one's share of the score.
3.	Opens the pin detail.	Shows the reports, photos and detection boxes.
Alternate Scenarios		
1a: A citizen token calls this route. The server answers 403. 2a: A photo could not be analysed. The pin is flagged and uses its category's default severity.		
Post Conditions		
Step#	**Description**	
1.	No data has changed unless the admin continues to UC-12 or UC-14.	
Use Case Cross referenced	UC-12, UC-13, UC-14	

4.2.12. UC-12: Split Merged Pin

An administrator corrects an incorrect automatic merge. No data is lost, because reports remain separate records attached to a pin.

UC-12: Split Merged Pin		
Use case Id:	UC-12	
Actors: Admin		
Feature: Duplicate merging		
Pre-condition:	The pin has two or more reports.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the pin and selects the reports that belong elsewhere.	Shows each report's merge decision: candidates, distances and thresholds.
2.	Confirms the split.	Creates a second pin in one transaction and moves the selected reports.
3.	(System step)	Recounts supporters on both pins, recomputes scores and writes an audit row.
Alternate Scenarios		
2a: All reports or none are selected. The split is refused.		
Post Conditions		
Step#	**Description**	
1.	Two pins exist and every original report is intact.	
2.	The audit log names the admin, the time and both states.	
Use Case Cross referenced	UC-11, UC-14	

4.2.13. UC-13: Decide Repair Claim

This use case implements Gate 5. A human decision is required because cooperating accounts could satisfy every automatic check.

UC-13: Decide Repair Claim		
Use case Id:	UC-13	
Actors: Admin		
Feature: Repair verification		
Pre-condition:	A claim is awaiting a decision.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the claim list and selects a claim.	Shows before and after photos, matched keypoints and each gate's measurements.
2.	Approves the claim, with an optional note.	Closes the pin and writes an audit row in the same transaction.
3.	(System step)	Awards the repair points once, within the per-pin cap, and notifies the volunteer.
Alternate Scenarios		
2a: The admin rejects the claim. The pin stays open, no points are paid and the reason is logged. 2b: The admin approves a claim that an automatic gate rejected. This is allowed and is logged as an override. 3a: The cap has been reached. The claim closes and no further points are paid.		
Post Conditions		
Step#	**Description**	
1.	The claim has a final decision.	
2.	Points are in the ledger only if the claim was approved.	
Use Case Cross referenced	UC-10, UC-16	

4.2.14. UC-14: Override Status and Read Audit Log

An administrator manually corrects the status of a pin, and can later review all sensitive actions in the audit log.

UC-14: Override Status and Read Audit Log		
Use case Id:	UC-14	
Actors: Admin		
Feature: Administration		
Pre-condition:	The admin is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Selects a new status for a pin and writes a note.	Saves the change and an audit row in one transaction.
2.	Opens the audit log and applies filters.	Lists actor, action, target, before and after values, and time.
Alternate Scenarios		
1a: The status value is not allowed. The server answers 422 and nothing changes.		
Post Conditions		
Step#	**Description**	
1.	The audit log holds the override.	
2.	Audit rows are never edited through the system.	
Use Case Cross referenced	UC-11, UC-12, UC-13	

4.2.15. UC-15: Receive Weather Warning

Citizens in a zone are warned before forecast rain makes road damage harder to see, and rain-sensitive hazards are ranked higher.

UC-15: Receive Weather Warning		
Use case Id:	UC-15	
Actors: Scheduler (time, initiator); Open-Meteo and Expo push service (secondary); Citizen		
Feature: Weather and ranking		
Pre-condition:	Zones are defined and the phone has shared its current zone while the app was open.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	(System step)	The scheduler fetches the forecast for each zone every hour and stores a snapshot.
2.	(System step)	When the risk value crosses the threshold, it logs the crossing time and recomputes scores in that zone.
3.	(System step)	Sends one push message to devices whose last known zone matches.
4.	Sees the warning on the phone.	The app reports the arrival time so that delivery delay can be measured.
Alternate Scenarios		
1a: Open-Meteo cannot be reached. The last snapshot stays in use and the failure is logged. 2a: A warning for the same event has already gone out. No repeat is sent.		
Post Conditions		
Step#	**Description**	
1.	A notification record holds the sent and received times.	
2.	Pin scores reflect the new risk.	
Use Case Cross referenced	UC-07, UC-11	

4.2.16. UC-16: View Points and Leaderboard

The citizen views a balance, its history and the leaderboard. The server calculates all values.

UC-16: View Points and Leaderboard		
Use case Id:	UC-16	
Actors: Citizen		
Feature: Missions and rewards		
Pre-condition:	The citizen is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the points screen.	Sums the ledger and returns the balance with its history.
2.	Opens the leaderboard.	Orders users by points. A tie goes to whoever reached the total first.
Alternate Scenarios		
1a: Report points are still held. They are shown as pending until the pin is corroborated or approved.		
Post Conditions		
Step#	**Description**	
1.	No data has changed.	
Use Case Cross referenced	UC-08, UC-10, UC-13, UC-17	

4.2.17. UC-17: Claim Coupon

The citizen exchanges points for a coupon and receives a unique code.

UC-17: Claim Coupon		
Use case Id:	UC-17	
Actors: Citizen		
Feature: Missions and rewards		
Pre-condition:	The citizen is logged in and an active coupon is in stock.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the coupon list and chooses one.	Shows the cost, the shop and whether the coupon is simulated.
2.	Taps Claim.	Locks the user row, checks the balance and stock, then writes a spending entry and an issued code in one transaction.
Alternate Scenarios		
2a: The balance or the stock is too low. The server answers 409 and nothing changes.		
Post Conditions		
Step#	**Description**	
1.	The balance is lower by the cost.	
2.	An issued code belongs to the citizen.	
Use Case Cross referenced	UC-16, UC-18	

4.2.18. UC-18: Redeem Coupon

An issued code is redeemed once. In the absence of a shop partner, an administrator screen performs this role.

UC-18: Redeem Coupon		
Use case Id:	UC-18	
Actors: Admin (acting as shop)		
Feature: Missions and rewards		
Pre-condition:	The citizen holds an issued code.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Enters the code.	Runs one conditional update that succeeds only if the code is still issued.
2.	(System step)	Confirms the redemption and records the time.
Alternate Scenarios		
1a: Two devices try at the same moment. Exactly one succeeds and the other receives 409. 1b: The code does not exist. The server reports that it was not found.		
Post Conditions		
Step#	**Description**	
1.	The code is permanently redeemed.	
Use Case Cross referenced	UC-17, UC-19	

4.2.19. UC-19: Manage Coupons

An administrator maintains the coupon catalogue.

UC-19: Manage Coupons		
Use case Id:	UC-19	
Actors: Admin		
Feature: Administration		
Pre-condition:	The admin is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Enters the title, shop, cost, stock and simulated flag.	Validates and saves the coupon and writes an audit row.
2.	Deactivates a coupon.	Hides it from the catalogue. Codes already issued stay valid.
Alternate Scenarios		
1a: The cost or stock is negative. The server answers 422.		
Post Conditions		
Step#	**Description**	
1.	The catalogue matches the admin's changes.	
2.	The change appears in the audit log.	
Use Case Cross referenced	UC-17, UC-18	

A user who has forgotten the password chooses a new one using a code sent to the account's email.

UC-20: Reset Forgotten Password		
Use case Id:	UC-20	
Actors: Citizen or Admin (initiator), SMTP mail server		
Feature: Accounts		
Pre-condition:	An account exists for the email address.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Taps Forgot password? on the login screen and enters the email.	Counts the request per email and network address, and gives the same answer whether or not the email is registered.
2.	(System step)	If the account exists, stores the keyed hash of a new 6-digit reset code and emails the code.
3.	Enters the code and the new password twice.	Checks the password rule on the phone and ticks each rule as it is met.
4.	Taps Set new password.	Checks the code's expiry and attempts, saves the new Argon2 hash, marks the email verified and ends all of the account's sessions.
Alternate Scenarios		
1a: A fourth request for the same email within ten minutes receives 429. 3a: The new password breaks the rule in FR-32. The server answers 422 with the reason. 4a: The code is wrong, expired or locked. The rules of UC-02 apply.		
Post Conditions		
Step#	**Description**	
1.	The account has the new password.	
2.	Every earlier session is revoked, so other phones must log in again.	
Use Case Cross referenced	UC-02, UC-03, UC-21	

A logged-in user replaces the password, for example after sharing a phone.

UC-21: Change Password		
Use case Id:	UC-21	
Actors: Citizen or Admin		
Feature: Accounts		
Pre-condition:	The user is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens Change password from the profile and enters the current password and the new one twice.	Ticks each password rule as it is met.
2.	Taps Change password.	Verifies the current password, checks the new one against FR-32, saves it and ends every other session of the account.
3.	(System step)	Keeps this phone logged in and confirms the change.
Alternate Scenarios		
2a: The current password is wrong. The server answers 400 and nothing changes. 2b: The new password equals the current one. The server answers 422. 2c: A sixth attempt within ten minutes receives 429.		
Post Conditions		
Step#	**Description**	
1.	The new password is in effect.	
2.	Only the session that made the change remains active.	
Use Case Cross referenced	UC-03, UC-04, UC-20	

A user ends the session on this phone.

UC-22: Log Out		
Use case Id:	UC-22	
Actors: Citizen or Admin		
Feature: Accounts		
Pre-condition:	The user is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Taps Log out on the profile screen.	Sends the refresh token, and the server revokes its session.
2.	(System step)	Deletes the tokens and the cached profile from the phone and opens the login screen.
Alternate Scenarios		
1a: There is no signal. The phone still logs out locally, and the session expires on the server within 14 days.		
Post Conditions		
Step#	**Description**	
1.	The session's refresh token is refused from now on.	
2.	Reports still queued on the phone stay queued for the same account's next login.	
Use Case Cross referenced	UC-03	

The citizen follows each report through to repair and sees totals of reports, confirmations and fixed hazards.

UC-23: View My Reports and Impact		
Use case Id:	UC-23	
Actors: Citizen		
Feature: Hazard reporting		
Pre-condition:	The citizen is logged in.	
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Opens the home screen.	Shows reports still queued on the phone and requests the citizen's reports and totals.
2.	(System step)	Returns each report with its state and its hazard's status, and the totals of reports, hazards, confirmations by others, hazards in repair and hazards fixed.
3.	Pulls the list down.	Retries queued uploads at once and refreshes the list and totals.
4.	Opens the profile.	Shows the rank and badges derived from the totals.
Alternate Scenarios		
1a: The server cannot be reached. Queued reports are still shown, and the message names the server address with a link to change it. 2a: A report was rejected. It is labelled Not accepted and is left out of the totals.		
Post Conditions		
Step#	**Description**	
1.	No data has changed, apart from queued reports that uploaded.	
Use Case Cross referenced	UC-05, UC-06, UC-16	

5. Non-functional Requirements

5.1. Performance Requirements

The approved proposal does not set pass or fail thresholds for performance. The values in Table 9 are therefore design targets to be measured during testing, not results already achieved.

Table 9: Performance requirements

ID	Requirement
NFR-01	95% of API requests, including the merge lookup, shall complete within 1 second with 20 concurrent users on the demonstration laptop and the seeded dataset.
NFR-02	The merge lookup shall use the spatial index and shall not scan all stored reports.
NFR-03	YOLO inference shall not run within a user request. On the demonstration laptop's CPU, the median analysis time shall be under 2 seconds and the 95th percentile under 4 seconds per image.
NFR-04	A confirmed report shall survive application restarts and repeated uploads without loss or duplication.
NFR-05	Forecasts shall refresh hourly, and warning delay shall be reported as a measured median.
NFR-06	Uploads shall be limited to 5 MB, and stored photographs to 1600 pixels on the long side.
NFR-07	The complete server stack shall start with a single command.

5.2. Safety Requirements

Table 10: Safety requirements

ID	Requirement
NFR-08	The application shall offer only tasks that a private person can lawfully and safely perform, and shall display safety guidance before a mission starts.
NFR-09	The application shall not require use while driving. A report needs a deliberate photograph and at most three taps on the report screen (shutter, category and Send), and warnings contain no instruction to act.
NFR-10	An incorrect merge shall be reversible, and merging shall never delete an original report.
NFR-11	No reward shall result from an automatic decision alone, so a software fault cannot release a payout without human approval.
NFR-12	A failure of the model or the weather service shall affect only that function, and reports shall still be accepted.

No safety certification applies to this academic prototype.

5.3. Security Requirements

Table 11: Security requirements

ID	Requirement
NFR-13	The server shall be the sole authority. Client values are treated as claims, and no point value is accepted from a client.
NFR-14	Passwords shall be hashed with Argon2, and verification codes and mission tokens stored only as hashes.
NFR-15	Rate limits shall apply per account and per device; for example, a sixth login attempt within one minute receives status 429.
NFR-16	Exact coordinates shall be available only to the reporter and administrators, and public views shall not identify a reporter.
NFR-17	Location shall be read only while the application is open, and a phone shall share only its zone identifier for weather targeting.

ID	Requirement
NFR-18	Uploads shall be treated as data: the type is determined from file content, and files are re-encoded and served from a separate origin.
NFR-19	User text shall be escaped on output, and the dashboard shall not render raw HTML from stored fields.
NFR-20	Secrets and raw photographs shall not be committed to the source repository.
NFR-21	Each sensitive administrator action shall be traceable to a person and a time.

5.4. User Documentation

A citizen guide covering registration, reporting, confirmation and coupons.

A reviewer guide covering the queue, pin splitting, claim review and the audit log.

Setup instructions in the repository, and API documentation generated at the /docs address.

An evaluation guide explaining how the figures for objectives O1 to O6 are regenerated.

6. References

[1] Provincial Disaster Management Authority, Government of Sindh, Flood 2022 in Sindh, from June 20, 2022 to March 31, 2023, p. 17. [Online]. Available: https://pdma.gos.pk/Documents/Reports/Flood%202022%20In%20Sindh.pdf (accessed Sep. 3, 2026).

[2] T. Siddiqui and I. Ayub, "Murad allocates Rs1.5bn for repair of few select rain-damaged roads in city," Dawn, Sep. 19, 2024. [Online]. Available: https://www.dawn.com/news/1859688 (accessed Nov. 7, 2026).

[3] A. Gardezi, "Govt celebrates Citizen Portal success despite only 30% satisfaction rate," ProPakistani, Aug. 6, 2026. [Online]. Available: https://propakistani.pk/2026/08/06/govt-celebrates-citizen-portal-success-despite-only-30-satisfaction-rate/ (accessed Nov. 7, 2026).

[4] mySociety, "FixMyStreet.com: The UK's street fault reporting website." [Online]. Available: https://www.mysociety.org/community/fixmystreet-in-the-uk/ (accessed Nov. 7, 2026).

[5] D. Arya, H. Maeda, S. K. Ghosh, D. Toshniwal, and Y. Sekimoto, "RDD2022: A multi-national image dataset for automatic road damage detection," arXiv preprint arXiv:2209.08538, 2022, doi: 10.48550/arXiv.2209.08538.

[6] National Coordination Office for Space-Based Positioning, Navigation, and Timing, "GPS Accuracy." [Online]. Available: https://www.gps.gov/gps-accuracy (accessed Sep. 4, 2026).

[7] PostGIS Project, "ST_ClusterDBSCAN," PostGIS Reference. [Online]. Available: https://postgis.net/docs/ST_ClusterDBSCAN.html (accessed Sep. 4, 2026).

[8] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in Proc. IEEE Int. Conf. Computer Vision (ICCV), 2011, pp. 2564-2571, doi: 10.1109/ICCV.2011.6126544.

[9] Ultralytics, "Ultralytics YOLO Docs." [Online]. Available: https://docs.ultralytics.com/ (accessed Sep. 4, 2026).

[10] Open-Meteo, "Weather API Features." [Online]. Available: https://open-meteo.com/en/features (accessed Sep. 4, 2026).

[11] Expo, "Location: LocationObject," Expo SDK Documentation. [Online]. Available: https://docs.expo.dev/versions/latest/sdk/location/ (accessed Sep. 4, 2026).

[12] A. Kumar, Sarichand, and R. Bai, "CivicPulse: A civic platform that proves a road hazard was actually repaired," Final Year Project Proposal 23F-90, Dept. of Computer Science, Sukkur IBA University, 2026.

[13] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD-96), 1996, pp. 226-231.

[14] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 779-788.

7. Appendices

Appendix A: Traceability

Table 12: Traceability of objectives to requirements and use cases

Objective	Requirements	Use cases
O1	FR-11 to FR-14	UC-05, UC-06, UC-12
O2	FR-15, FR-16	UC-05, UC-11
O3	FR-21 to FR-24	UC-10, UC-13
O4	FR-19, FR-20	UC-11
O5	FR-17 to FR-19	UC-15
O6	FR-25 to FR-29	UC-08, UC-09, UC-16, UC-17, UC-18
Common base	FR-01 to FR-10, FR-30 to FR-38	UC-01 to UC-04, UC-07, UC-14, UC-19 to UC-23

Table 13 traces each functional requirement to the test cases that check it, listed in Table 8 of the Software Design Specification.

Table 13: Functional requirements traced to test cases

Requirement	Test cases (SDS Table 8)
FR-01	TC-01
FR-02	TC-02, TC-03
FR-03	TC-07, TC-46, TC-47
FR-04	TC-04, TC-06
FR-05	TC-48
FR-06	TC-49
FR-07	TC-50
FR-08	TC-08, TC-09, TC-10
FR-09	TC-11 to TC-14
FR-10	TC-45
FR-11	TC-17, TC-19, TC-20
FR-12	TC-15, TC-16
FR-13	TC-18
FR-14	TC-21
FR-15	TC-23, TC-24
FR-16	TC-23, TC-25
FR-17	TC-30
FR-18	TC-29
FR-19	TC-25 to TC-28
FR-20	TC-51
FR-21	TC-36
FR-22	TC-31 to TC-34
FR-23	TC-37, TC-61
FR-24	TC-35
FR-25	TC-38, TC-39
FR-26	TC-40, TC-52
FR-27	TC-41
FR-28	TC-42
FR-29	TC-53

Requirement	Test cases (SDS Table 8)
FR-30	TC-54
FR-31	TC-44
FR-32	TC-55
FR-33	TC-56
FR-34	TC-57
FR-35	TC-58
FR-36	TC-59
FR-37	TC-59
FR-38	TC-60

Appendix B: Initial Parameter Values

These values are starting points. They remain subject to supervisor confirmation and tuning from measured data.

Table 14: Initial parameter values

Item	Initial value
Pair threshold	min(ceiling, base + k x mean accuracy), with base 5 m, k = 1.0 and ceiling 30 m
Active window	60 days
Severity	Based on the share of the image covered by detection boxes: below 1% gives 1, below 3% gives 2, below 7% gives 3, below 15% gives 4, and otherwise 5. Three or more defects add 1, to a maximum of 5.
Priority score	100 x (0.40 S + 0.25 C + 0.20 T + 0.15 W), where S = severity / 5, C = min(1, ln(1 + n) / ln(11)) for n supporters, T = min(1, days open / 30), and W = zone risk x category sensitivity
Weather risk	Forecast rainfall for the next six hours / 25 mm, capped at 1. A warning is issued at a risk of 0.5 or above with a probability of at least 60%.
Points	New report 10, confirmation 5, volunteer task 20, repair photograph 30; per-pin cap 100